

MA6224: Unmanned Vehicles

Course Project Specification

Nonlinear State Estimation (EKF) and Constrained Trajectory Tracking (NMPC) for
an Autonomous Quadrotor

School of Mechanical and Aerospace Engineering

Academic Year 2026–2027

Project Metadata

Course Code: MA 6224 — Unmanned Vehicles

Project Weightage: 20% of Total Course Grade

Deliverables: Technical Report (IEEE format) with Validated Source Code and Presentation

Collaboration: Groups of three (3) students

1. Executive Summary & Motivation

Autonomous multirotor aerial platforms operating in cluttered or turbulent environments require agile, robust, and constraint-aware feedback control systems. Unlike ground robots, quadrotors are underactuated, highly coupled, and subject to hard input saturations (rotor thrust minimums/maximums) as well as critical state constraints (e.g., maximum bank angles to prevent vortex ring states or aerodynamic stall).

In this project, students will design, implement, and validate an integrated Guidance, Navigation, and Control (GNC) architecture for a 6-DOF quadrotor tracking an agile 3D **figure-8 (lemniscate)** trajectory. The project consists of two complementary components:

- 1. Nonlinear State Estimator (EKF):** An Extended Kalman Filter fusing high-rate noisy inertial measurements (IMU: 3-axis accelerometer and 3-axis gyroscope) with lower-rate positioning fixes (global navigation satellite system) to provide optimal full-state estimates $\hat{\mathbf{x}}$ and error covariance $\mathbf{P}$.
- 2. Nonlinear Model Predictive Controller (NMPC):** A receding-horizon optimal controller that natively incorporates the quadrotor's nonlinear translational and rotational dynamics, explicitly enforcing rotor thrust saturation limits and attitude tilt safety envelopes.

2. Quadrotor Kinematics and Dynamic Modeling

2.1. Coordinate Frames and State Definitions

We consider a world reference frame $\mathcal{F}_W = \{O_W, \mathbf{x}_W, \mathbf{y}_W, \mathbf{z}_W\}$ adhering to the North-East-Down (NED) convention, and a body-fixed frame $\mathcal{F}_B = \{O_B, \mathbf{x}_B, \mathbf{y}_B, \mathbf{z}_B\}$ located at the center of mass (CoM) of the vehicle.

The system is described by the 13-dimensional state vector $\mathbf{x} \in \mathbb{R}^{13}$: where:

- $\mathbf{p}^W = [x, y, z]^T \in \mathbb{R}^3$: Position coordinates of the CoM in the inertial frame (m).
- $\mathbf{v}^W = [u, v, w]^T \in \mathbb{R}^3$: Linear velocity vector in the inertial frame (m/s).

- $\boldsymbol{\eta}^{WB} = [\eta_w, \eta_x, \eta_y, \eta_z]^T \in \mathbb{R}^4$: A unit quaternions.
- $\boldsymbol{\omega}^B = [p, q, r]^T \in \mathbb{R}^3$: Angular body rates expressed in the body frame $\mathcal{F}_B$ (rad/s).

The collective thrust $\mathbf{T}_B$ and body torque $\boldsymbol{\tau}_B$ are generated from individual rotor thrusts T_i :

$$\mathbf{T}_B = \begin{bmatrix} 0 \\ 0 \\ \sum T_i \end{bmatrix}$$

$$\boldsymbol{\tau}_B = \begin{bmatrix} d_y(-T_0 - T_1 + T_2 + T_3) \\ d_x(-T_0 + T_1 + T_2 - T_3) \\ c_\tau(-T_0 + T_1 - T_2 + T_3) \end{bmatrix}$$

where d_x and d_y are the rotor displacements from the center of the mass of the quadrotor and c_τ is the rotor drag torque. constant.

2.2. Continuous Equations of Motion

The full nonlinear translational and rotational dynamics are governed by:

$$\dot{\mathbf{x}} = f_{dyn}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \mathbf{v}^W \\ \mathbf{q}^{WB} \begin{bmatrix} 0 \\ \frac{\boldsymbol{\omega}^B}{2} \end{bmatrix} \\ \mathbf{q}^{WB} \odot \mathbf{T}_B + \mathbf{g}^W \\ \mathbf{J}^{-1}(\boldsymbol{\tau}_B - \boldsymbol{\omega}^B \times \mathbf{J}\boldsymbol{\omega}^B) \end{bmatrix}$$

where $\mathbf{J} = \text{diag}(J_{xx}, J_{yy}, J_{zz})$ is the inertia matrix, the quadrotor's position in the World frame is given by $\mathbf{p}^W$, its attitude state of the body is given by $\boldsymbol{\eta}^{WB}$, its linear velocities in the World frame is denoted by $\mathbf{v}^W$, $\mathbf{g}^I = [0, 0, g]^T$ with $g = 9.81 \text{ m/s}^2$, and its angular velocity is given by $\boldsymbol{\omega}^B$.

2.3. Nominal Benchmark Parameters

Table 1 specifies the physical model parameters to be used in all student simulations.

Table 1: Nominal Physical and Operational Parameters of the Quadrotor

Parameter Description	Symbol	Nominal Value	Bounds / Range	Units
Total Mass	m	1.20	—	kg
Arm Lengths	d_x, d_y	0.225	—	m
Roll Moment of Inertia	J_{xx}	1.25×10^{-2}	—	$\text{kg} \cdot \text{m}^2$
Pitch Moment of Inertia	J_{yy}	1.25×10^{-2}	—	$\text{kg} \cdot \text{m}^2$
Yaw Moment of Inertia	J_{zz}	2.20×10^{-2}	—	$\text{kg} \cdot \text{m}^2$
Torque-to-Thrust Ratio	c_τ	0.015	—	m
Rotor Thrust Bound (per rotor)	f_i	—	[0.2, 5.5]	N
Total Collective Thrust Bound	T	11.77 (hover)	[0.8, 22.0]	N
Maximum Allowable Attitude Angles	ϕ, θ	0.00 (hover)	$[-35^\circ, +35^\circ]$	deg
Maximum Body Angular Rates	p, q, r	0.00	$[\pm 180, \pm 180, \pm 90]$	deg/s

3. Mission Profile: 3D Lemniscate (Figure-8) Trajectory

The reference path to be tracked is an agile 3D continuous parametric lemniscate defined over $t \in [0, t_f]$:

$$x_r(t) = A_x \sin(\omega_0 t) \quad (1)$$

$$y_r(t) = A_y \sin(2\omega_0 t) \quad (2)$$

$$z_r(t) = z_0 + A_z \cos(\omega_0 t) \quad (3)$$

$$\psi_r(t) = \text{atan2}(\dot{y}_r(t), \dot{x}_r(t)) \quad (4)$$

Trajectory Parameters:

- Amplitudes: $A_x = 3.0$ m, $A_y = 2.0$ m, $A_z = 0.5$ m.
- Nominal Flight Altitude: $z_0 = -2.5$ m (NED convention: negative z indicates elevation above ground).
- Angular Velocity Parameter: $\omega_0 = 0.25$ rad/s (Period $T_{\text{period}} = \frac{2\pi}{\omega_0} \approx 25.13$ s).
- Mission Duration: $t_f = 60$ s (completes approximately 2.4 full circuits).

If you require the reference linear velocities $\mathbf{v}_d(t) = [\dot{x}_d, \dot{y}_d, \dot{z}_d]^T$ and accelerations $\mathbf{a}_d(t) = [\ddot{x}_d, \ddot{y}_d, \ddot{z}_d]^T$, analytically compute to populate the reference trajectory vector $\mathbf{p}_r(t)$.

4. Sensor Characteristics

The controller receives estimated states $\hat{\mathbf{x}}$ from an Extended Kalman Filter that fuses high-rate inertial measurements with low-rate absolute positioning sensors.

1. Inertial Measurement Unit (IMU) — Rate: 100 Hz ($\Delta t_{\text{imu}} = 0.01$ s):

Standard deviations for accelerations: $\sigma_a = 0.08$ m/s², $\sigma_\omega = 0.015$ rad/s. Accelerometer bias $\mathbf{b}_a$ and gyroscope bias $\mathbf{b}_\omega$ exhibit random walk dynamics: $\dot{\mathbf{b}}_a = \mathbf{w}_{ba}$, $\dot{\mathbf{b}}_\omega = \mathbf{w}_{bg}$.

2. Positioning Sensor (GNSS) — Rate: 20 Hz ($\Delta t_{\text{pos}} = 0.05$ s):

Standard deviations for positions: $\sigma_p = 0.02$ m,

5. Recommended Solvers and Software Frameworks

To avoid writing custom interior-point optimizers from scratch, students are advised to use:

- **MATLAB:** CasADi (v3.5+) with IPOPT, the Model Predictive Control Toolbox (nlmpc object), or FORCESPRO.
- **Python:** CasADi with IPOPT, or acados using the Real-Time Iteration (RTI) Sequential Quadratic Programming (SQP) scheme.

6. Project Deliverables & Submission Guidelines

Each submission must include:

1. Technical Final Report (PDF, IEEE Conference Style, Max 6 Pages):

- Detailed explanation of controller and observer tuning strategies.
- Required plots: 3D trajectory tracking vs. reference; state tracking error history; EKF estimation errors with $\pm 2\sigma$ bounds; actuator thrust and torque time-histories verifying constraint boundaries.

- Critical discussion of solver computation times and execution limits.
2. **Reproducible Source Code:** A self-contained MATLAB or Python repository featuring a top-level execution script (`main_simulation.m` or `main_simulation.py`) and a descriptive `README.md`.
 3. **Flight Visualization Video:** A recorded MP4 video (maximum 2 minutes) displaying the quadrotor navigating the 3D figure-8 path alongside real-time error telemetry.
 4. **Project Presentation (Week 7):** 6-minute presentation followed by a brief Q&A session.

7. Detailed Evaluation Rubric

Rubric will be available in later date as an additional document.